

I. METHOD FOR SELECTING PARAMETERS

This note provides a Monte-Carlo method for searching the optimal parameter. This algorithm is less novel, so it is not included in our manuscript. Readers may also employ their own methods to determine the optimal parameters. Section I-A provides the search algorithm, while Section I-B lists the obtained optimal parameters.

The following points should be noted:

- 1) **Given a $\mathcal{P}(N, K)$ polar code, the optimal θ is not unique.**
- 2) **Polar codes with different lengths and rates may share the same optimal θ .**
- 3) **A theoretical expression exists that can directly compute the optimal θ .**
- 4) During the searching, the underlying decoder can be replaced with any polar decoder, such as the BP, SCL, and OSD decoding.
- 5) If the perturbation-based SCL decoding is considered, the optimal θ is always close to 0.1.
- 6) Our simulation results show that the perturbation-based decoding performs better with dynamic perturbation power than with a fixed perturbation power (see my work in ISIT-2025 and ITW-2025).
- 7) For some polar codes, the performance gap between the optimal value of θ and a near-optimal choice is typically very small, often less than 0.05 dB. From our perspective, it is unnecessary to insist on searching exact optimal θ .
- 8) All equation numbers referenced in this note correspond to those in our manuscript submitted to TCOM.

A. Selection Method

The selection of θ can be realized through the following steps:

- 1) Consider a $\mathcal{P}(N, K)$ polar code with the SC decoding. Determine the channel SNR β at which the FER performance reaches 10^{-4} .
- 2) Based on equation (22), compute the upper bound θ_{up} at SNR = β dB. If $\theta \leq \theta_{\text{up}}$, $P_{\text{sign}} \geq 0.95$; otherwise, $P_{\text{sign}} < 0.95$.
- 3) Search for the optimal value within $\theta \in (0, \theta_{\text{up}}]$. The search step is 0.1. Select the value of θ that minimizes the FER performance of the perturbation-based decoder, when decoding the $\mathcal{P}(N, K)$ polar code at an SNR of β dB.
- 4) Output θ .

Remark: Our previous researches show that overly small θ or overly larger θ provide little improvement for the perturbation-based decoding. Thus, the practical search range can be restricted to $\theta \in [0.1, \theta_{\text{up}}]$ and $\theta_{\text{up}} \in [0.1, 1]$.

Moreover, note that ε is chosen as the modulation magnitude. E.g., with BPSK modulation, $\varepsilon = 1$.

B. Empirical Parameters

For the $\mathcal{P}(N, K)$ polar code with rate of $1/2$, the optimal θ is approximately 0.3 when the block length is between 8192 and 1024, and it reduces to around 0.1 when the block length exceeds 16384. If the codeword length is less than 1024, it is better to set $\theta = 0.5$.